TEK-UP
Ecole Supérieure Privée
Technologie & Ingénierie

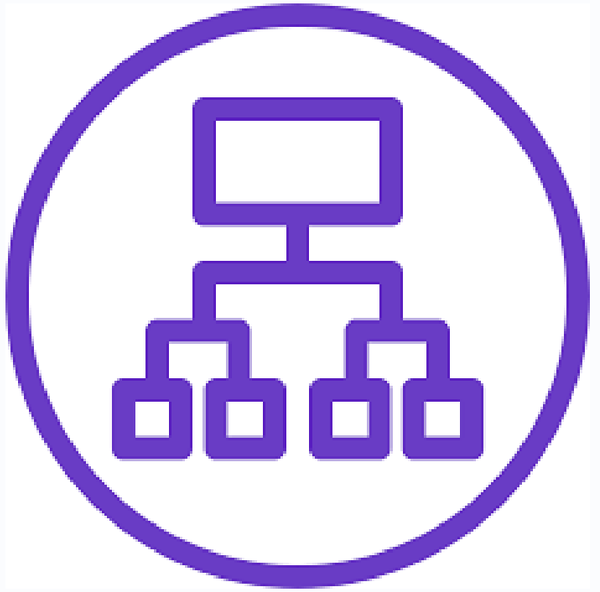
AWS LAB 01 EC2



By Alouani Mohamed

Made with GAMMA

Application Load Balancer



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Create EC2 Instances

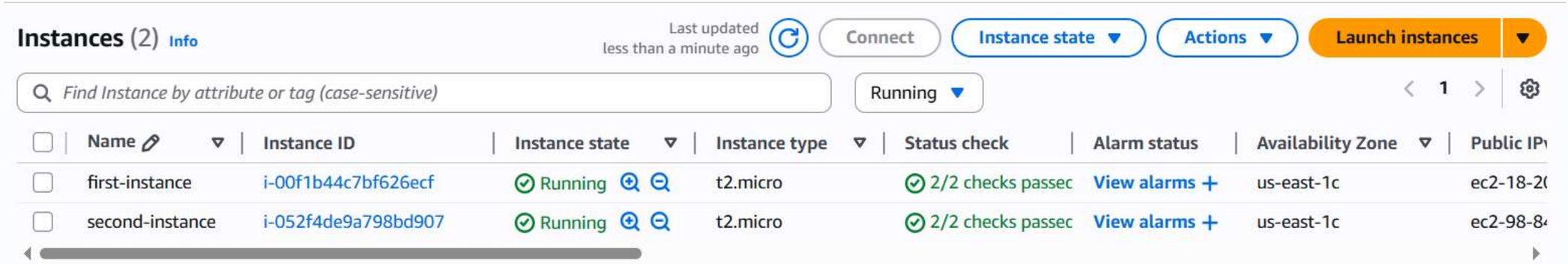
1. Créer deux instances EC2 : **first-instance** et **second-instance** :
2. Vous pouvez suivre le TP AWS-LAB-01-EC2 de la semaine passée
    - Utiliser le même **Security Group**.
    - Utiliser la même **Key Pair**.
3. Pour le **User Data Script** utiliser ce script pour afficher les le hostname de chaque instance :

```
#!/bin/bash
# Use this for your user data (script from top to bottom)
# install httpd (Linux 2 version)
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
echo "<h1>Hello World from $(hostname -f)</h1>" >
/var/www/html/index.html
```

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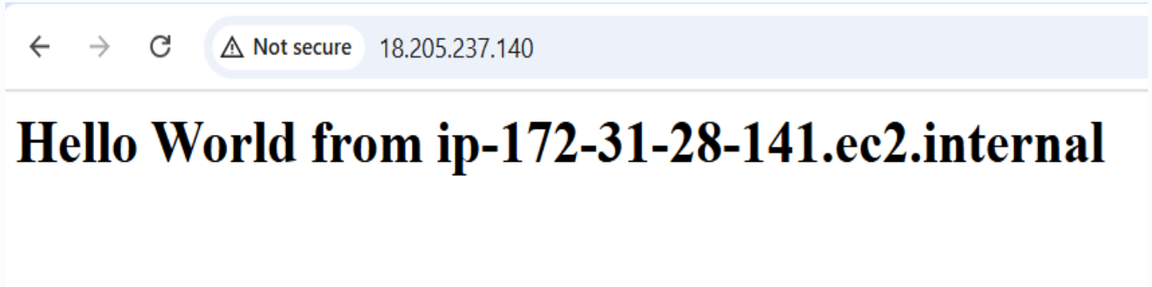
Create EC2 Instance

- You should see your two instances running :

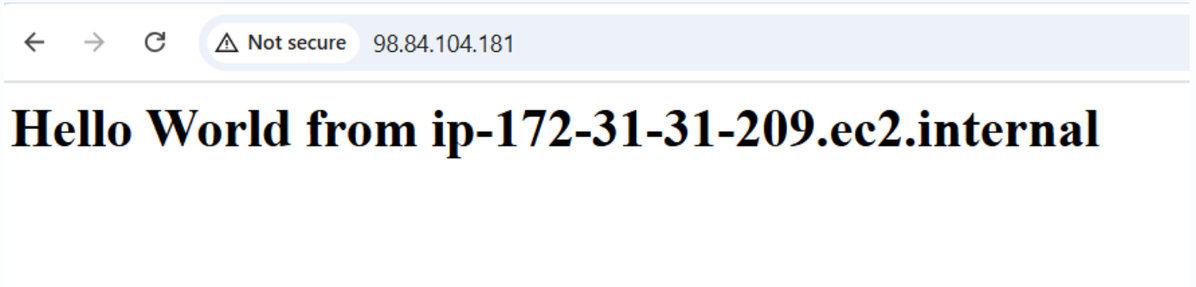


- and each instance should display its hostname on the web via its IPv4 address :

first-instance



second-instance

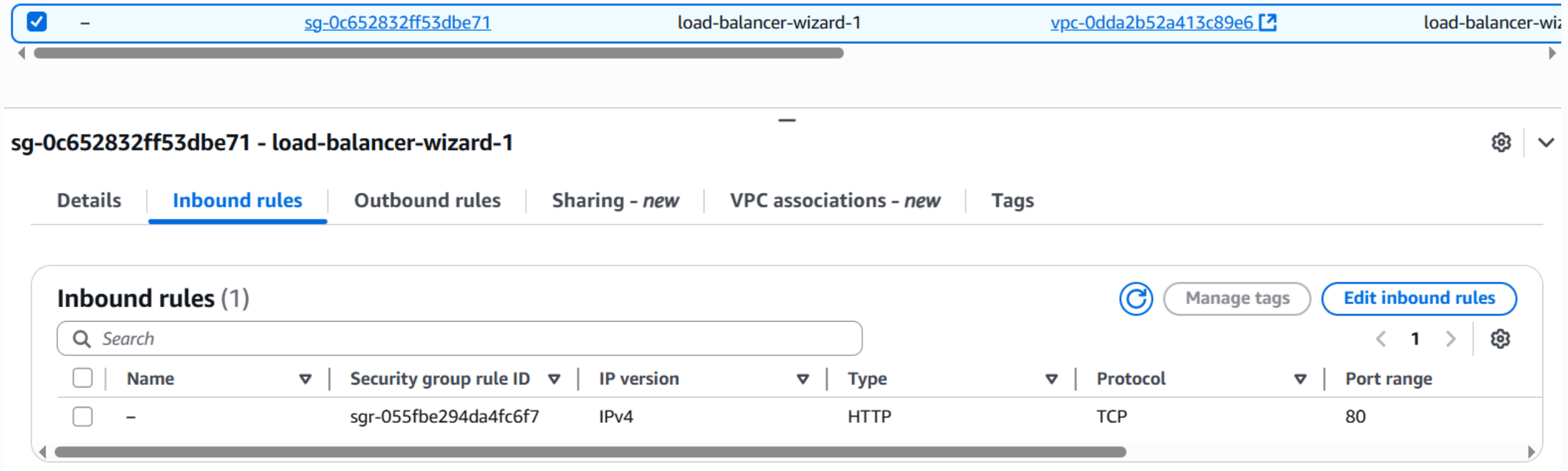


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Create Application Load Balancer

Let's create an Application Load Balancer to distribute traffic evenly across our instances.

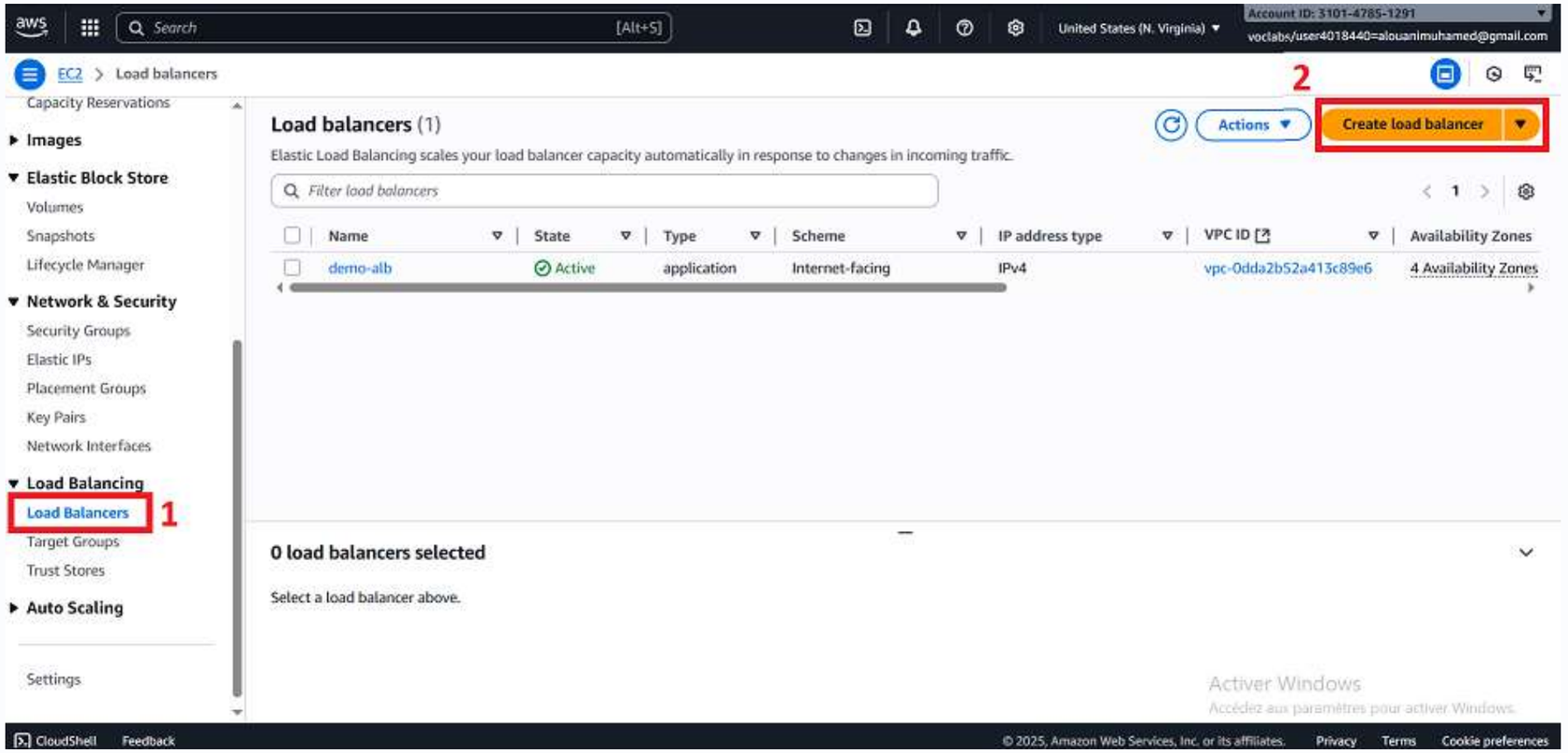
1. First, we need a security group for our load balancer. Create one named "load-balancer-wizard-1" that allows HTTP traffic on port 80.



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Create Application Load Balancer

2. Select In the sidebar, scroll down to **“Load Balancers”**
   then click **“Create Load Balancer”** in the top-right corner.

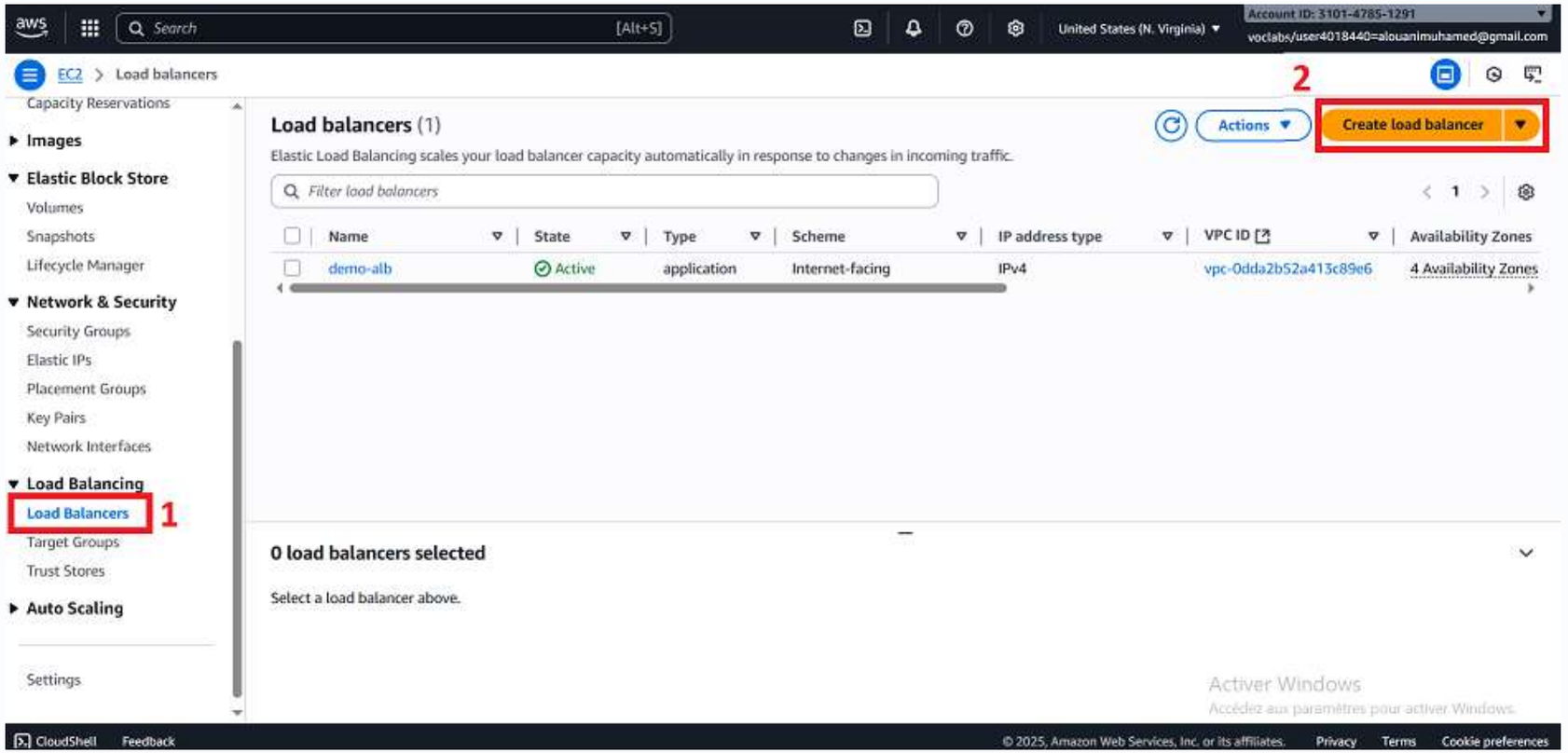


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Create Application Load Balancer

2. Select In the sidebar, scroll down to **“Load Balancers”**

   then click **“Create Load Balancer”** in the top-right corner.

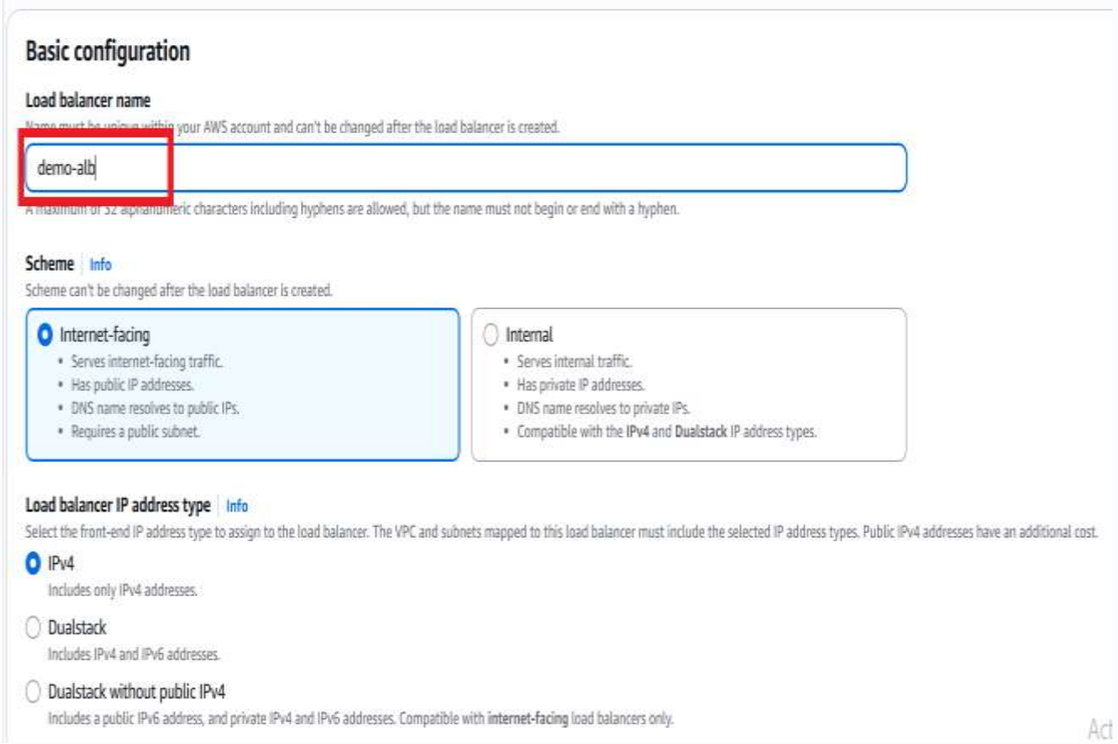


3. Choose **“Application Load balancer”**

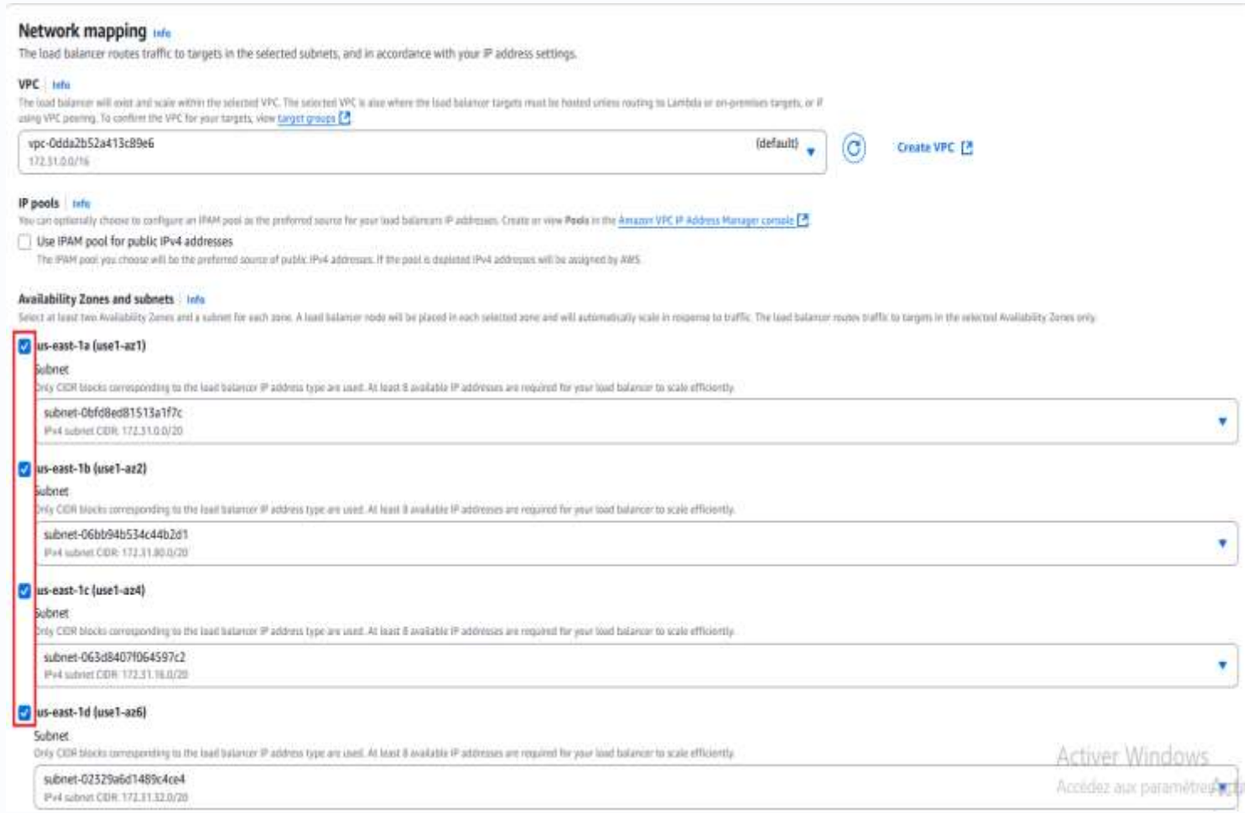
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Create Application Load Balancer

4. Set a name for our Load balancer



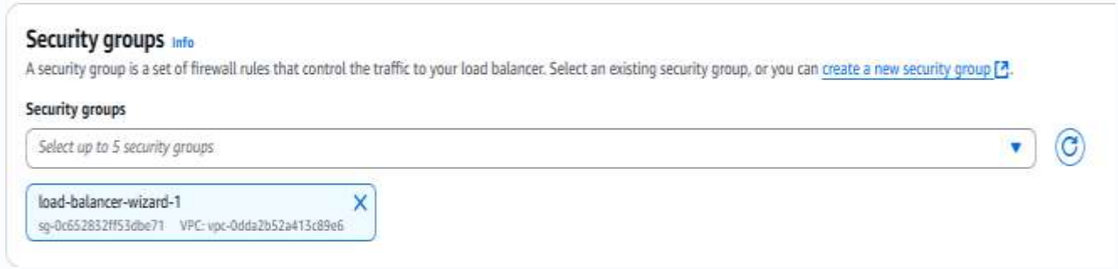
5. For network mapping, select 3 or 4 Availability Zones (verify that your instances are located within them) to ensure our load balancer is highly available.



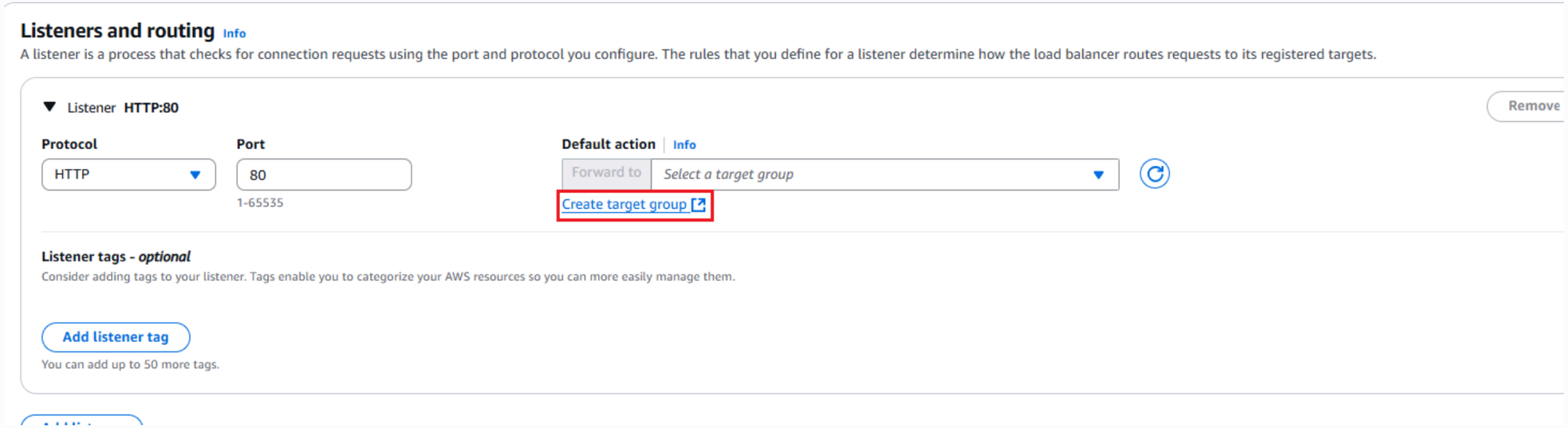
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Create Application Load Balancer

6. Select the security group we created for the load balancer.



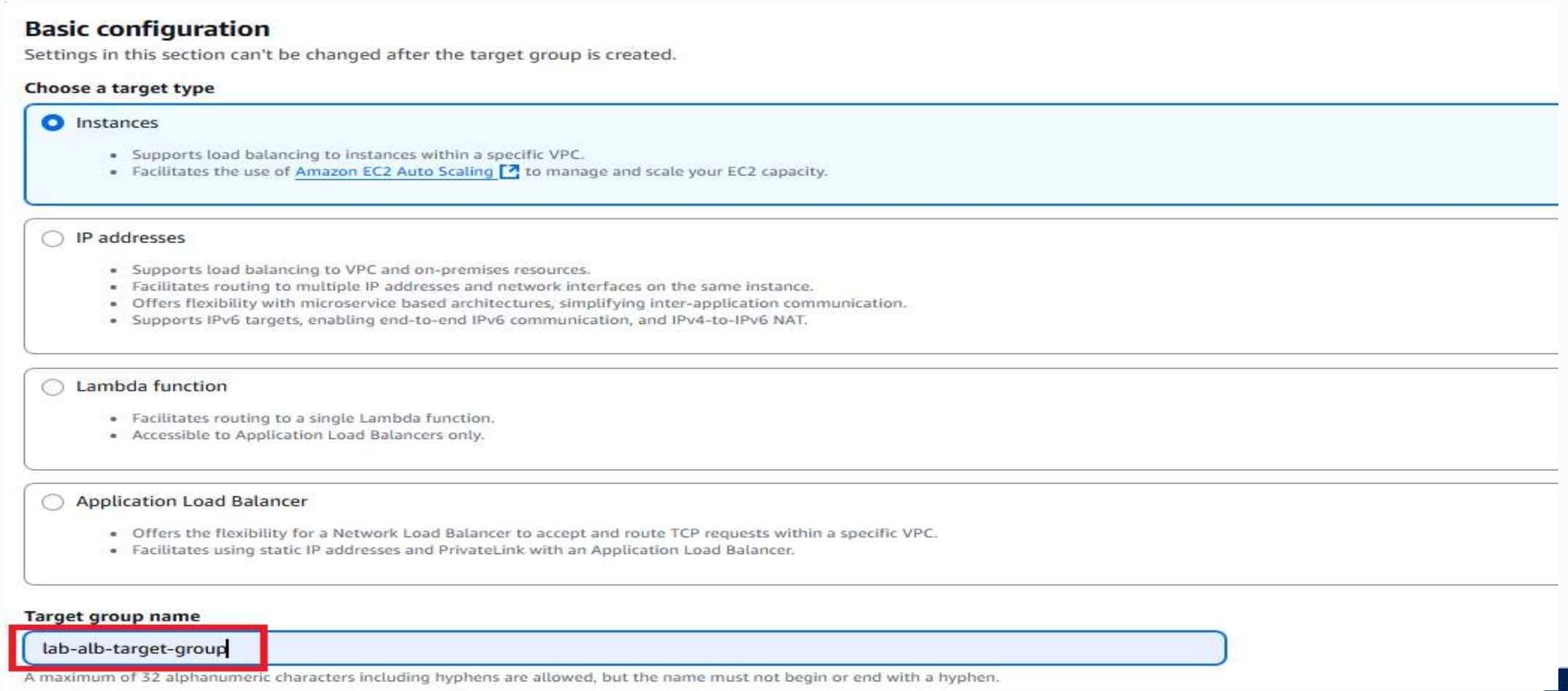
7. Select For Listeners and Routing, create a new target group.



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Create Application Load Balancer

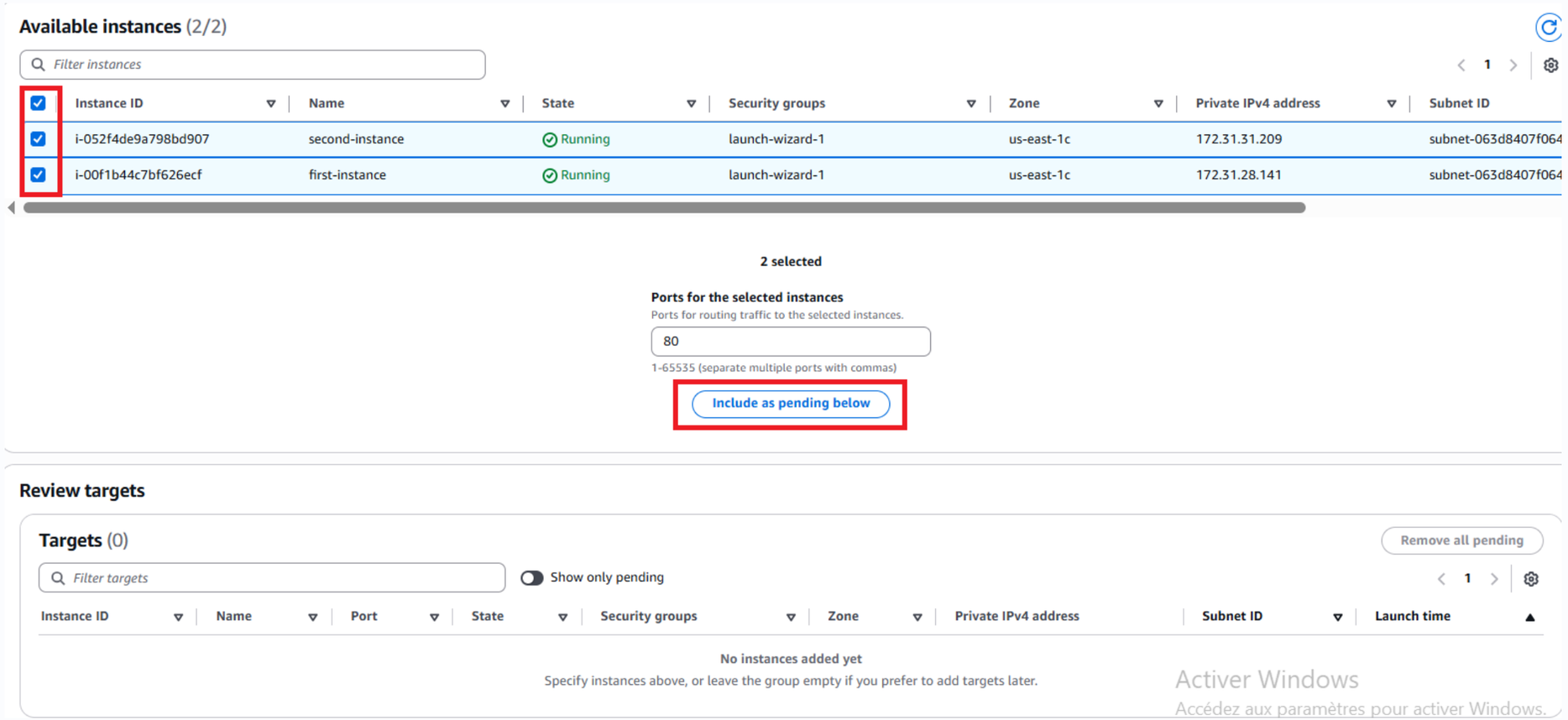
8. On the target group creation page, give it a name, leave all other settings as default, and click "Next”.



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Create Application Load Balancer

9. On Select the two instances in "**Available Instances“,** then click "**Include as Pending**" below. They will be added under "**Review Targets**".

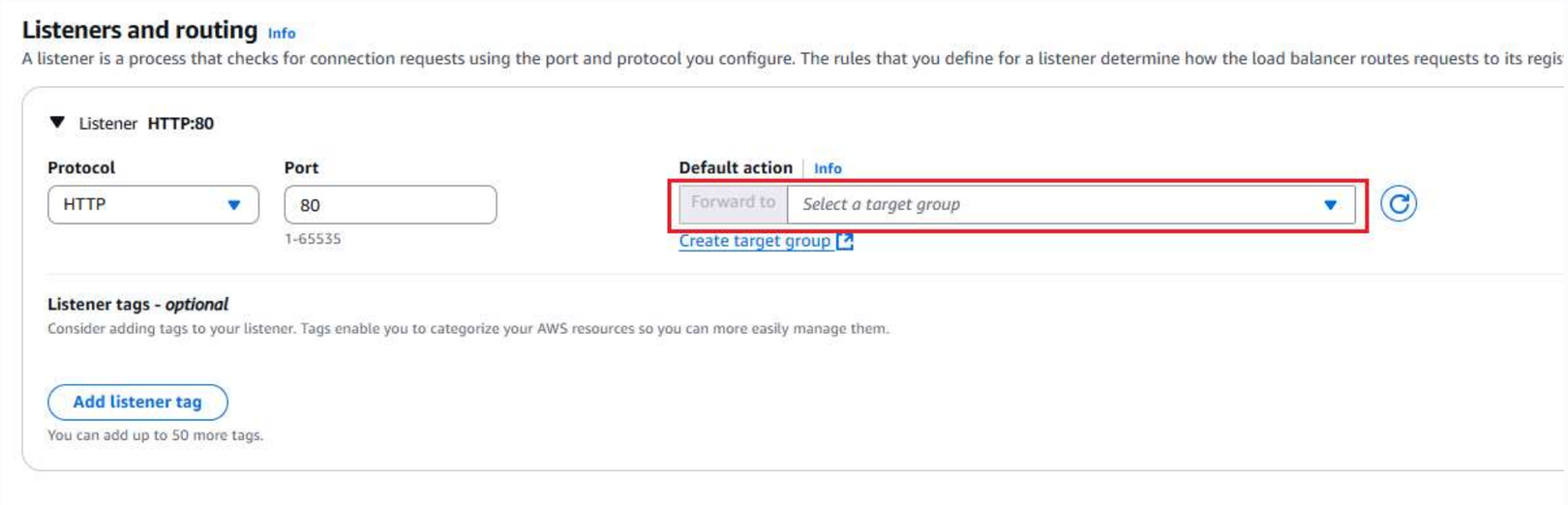


10. Click **“Create target group”**

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Create Application Load Balancer

11. Return to the load balancer creation page and select the target group we just created.

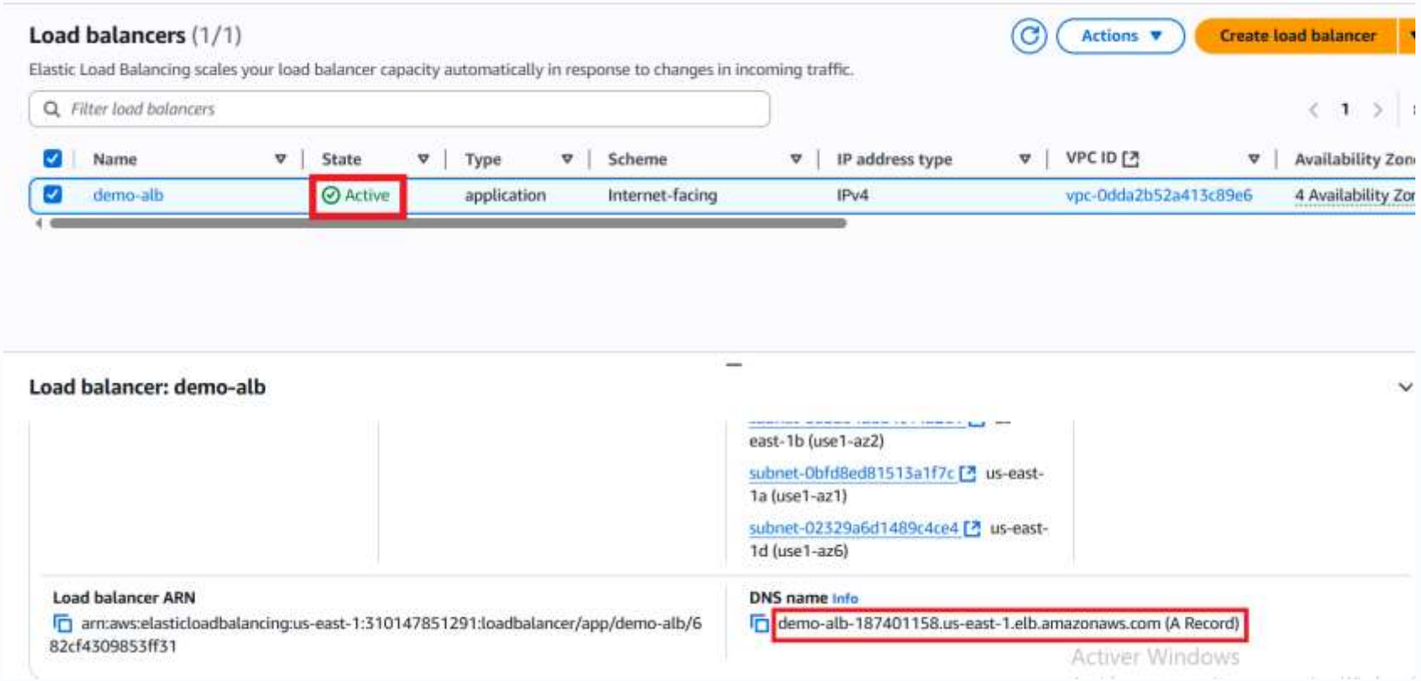


12. Click **"Create Load Balancer".** It will take a few minutes to become active, so please wait.

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Create Application Load Balancer

13. Once your load balancer is active, copy its DNS name, paste it into your browser, and reload the page.



⇨ You will see that the load balancer distributes traffic across our two instances.

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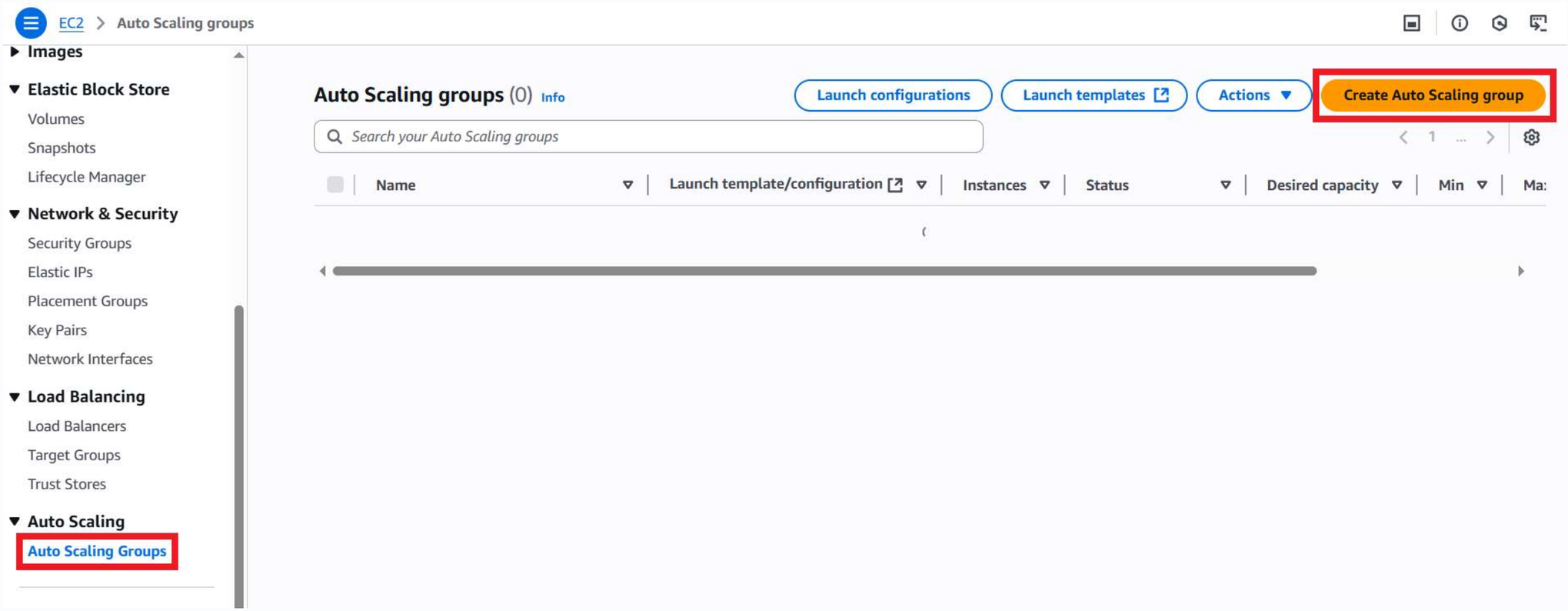
Auto-Scaling Group



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Create Auto-Scaling Group

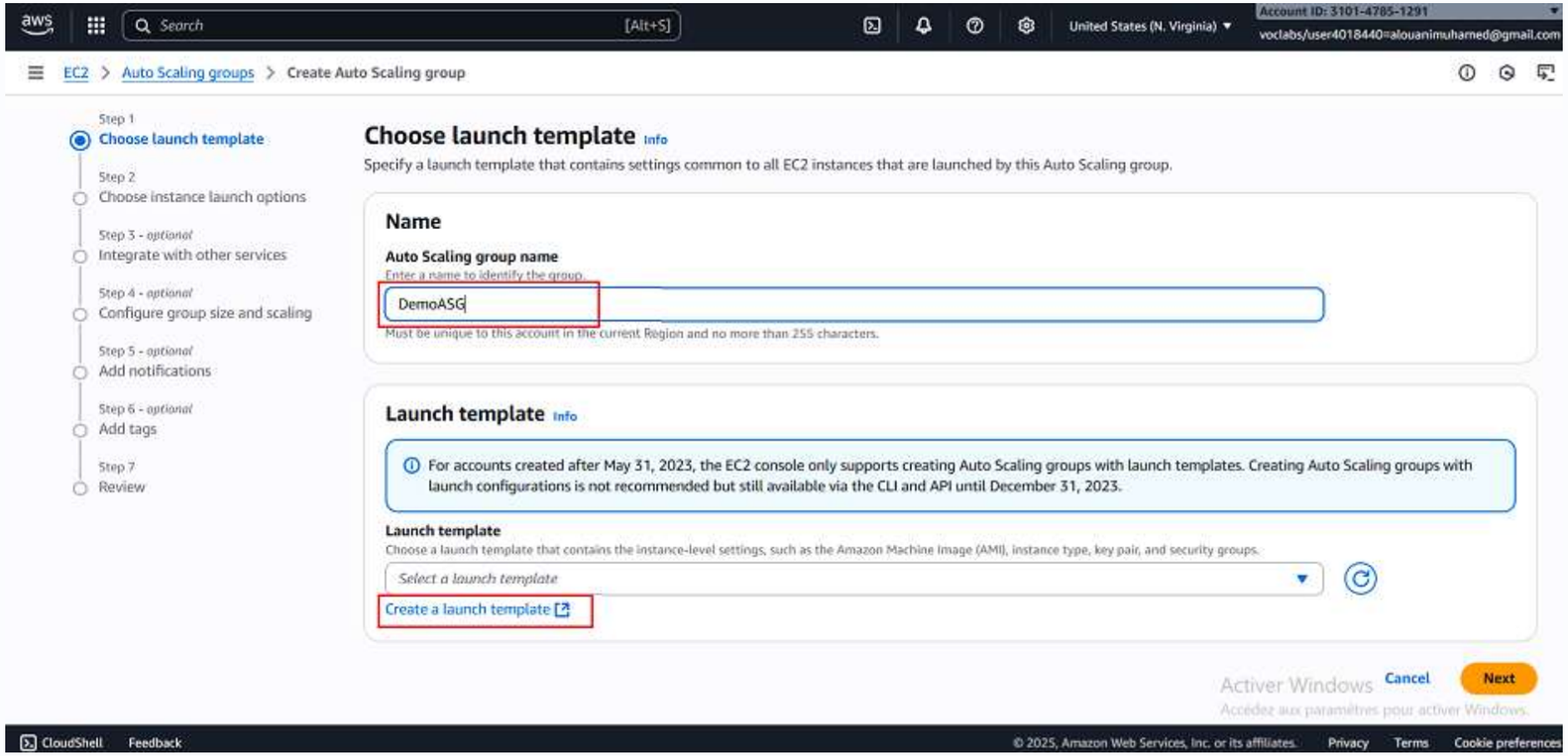
1. Terminate or stop the two instances: **first-instance** and **second-instance**.
2. In the sidebar, scroll down to **Auto Scaling Groups**, then click **“Create Auto Scaling Group”**.



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Create Auto-Scaling Group

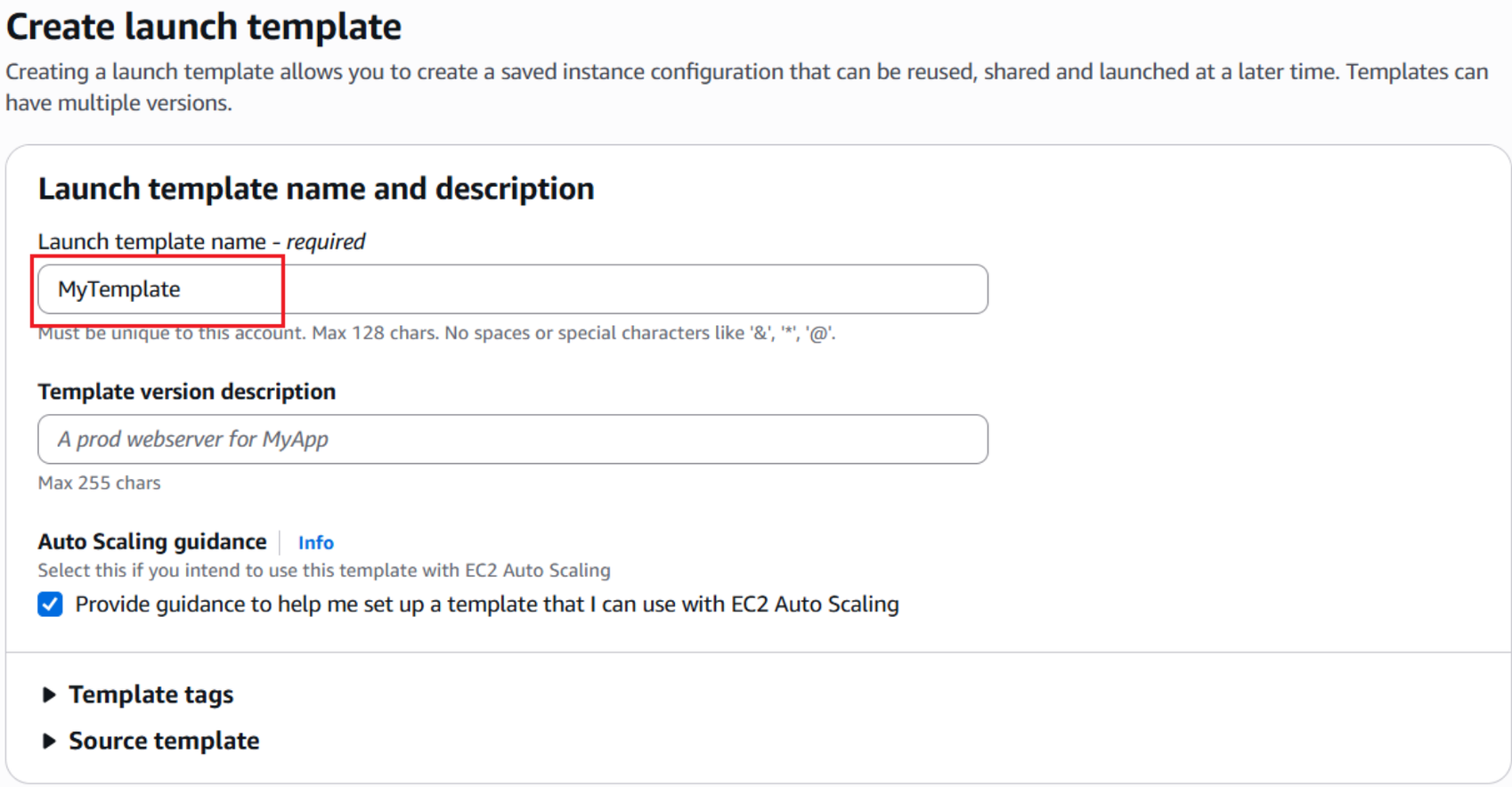
3. Give a name to your Auto Scaling Group (ASG).
4. Then, create a launch template.



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Create Launch Template

5. Give a name to your launch template

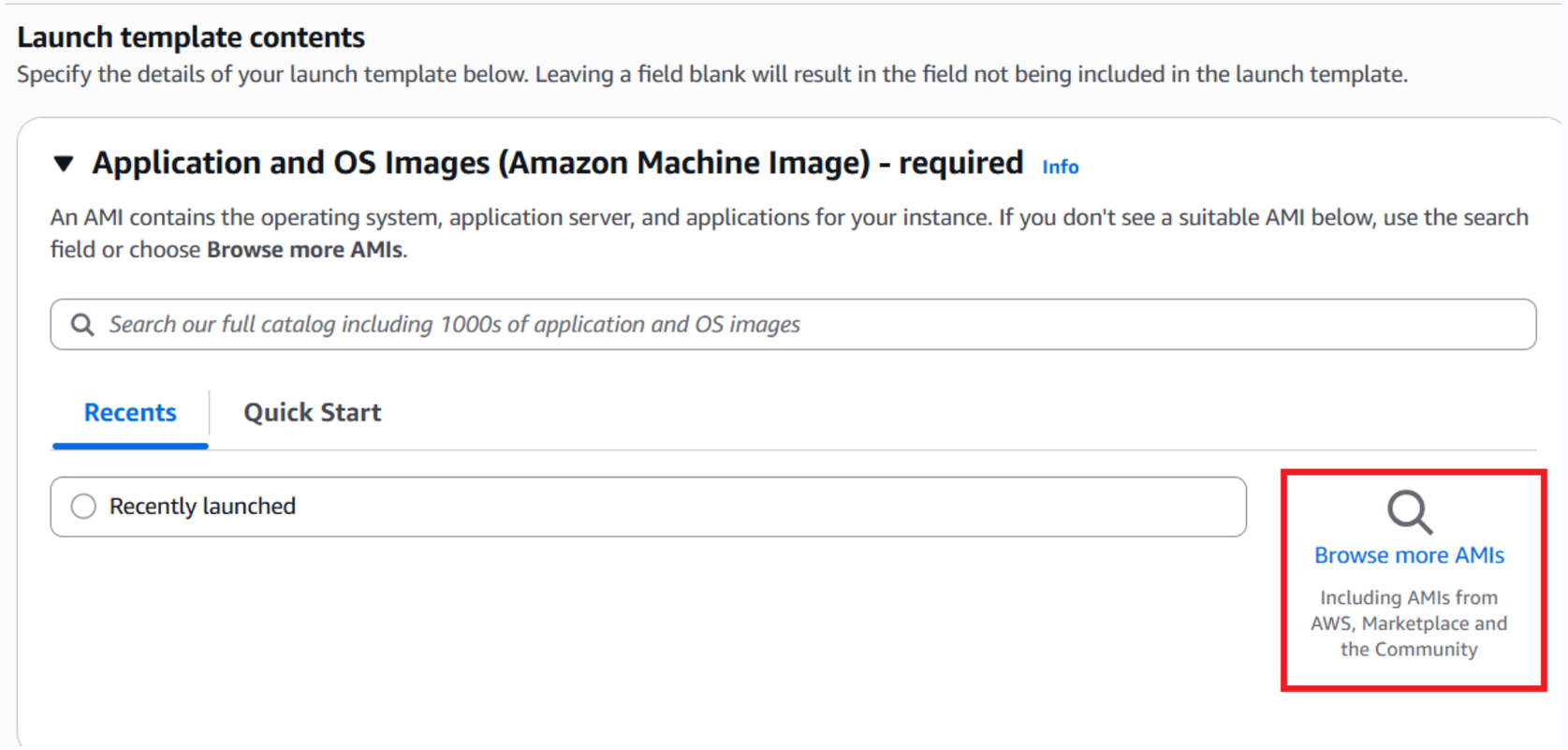


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Create Launch Template

6. Terminate For the AMI, we will use **Amazon Linux 2023**.

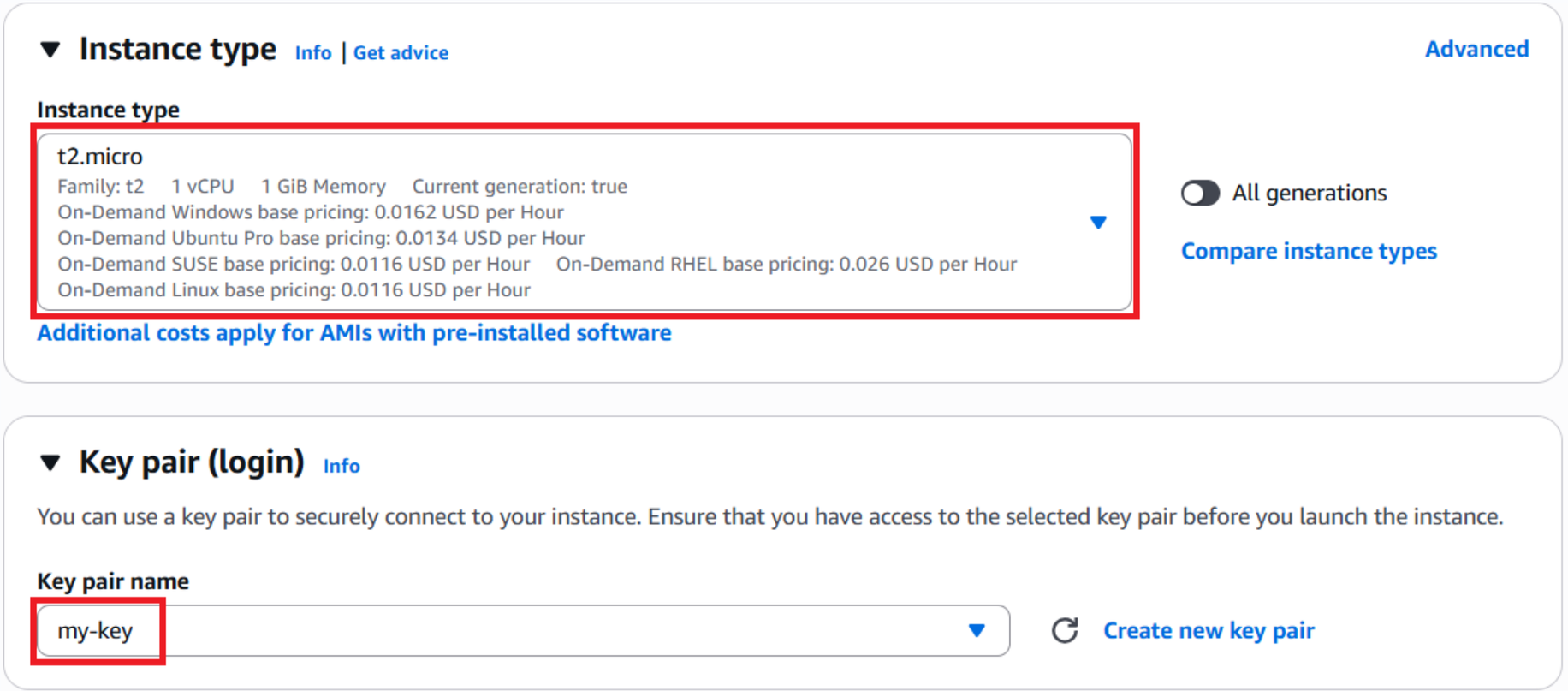
   Browse the list of AMIs and select the first one.



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Create Launch Template

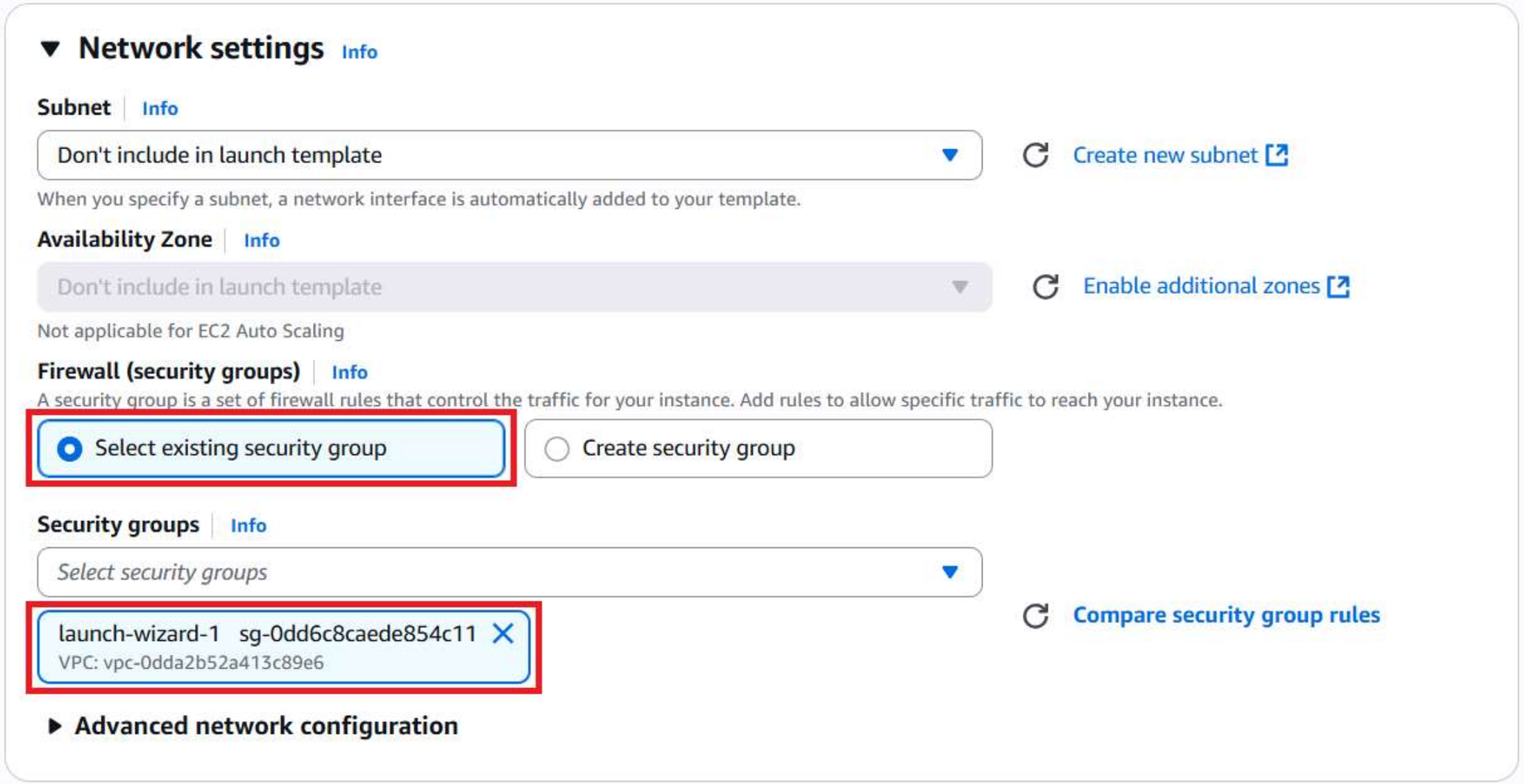
7. Choose “**t2.micro**” as the instance type.
8. Select the existing **key pair** (the one created in the **“AWS-LAB-01-EC2“**).



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Create Launch Template

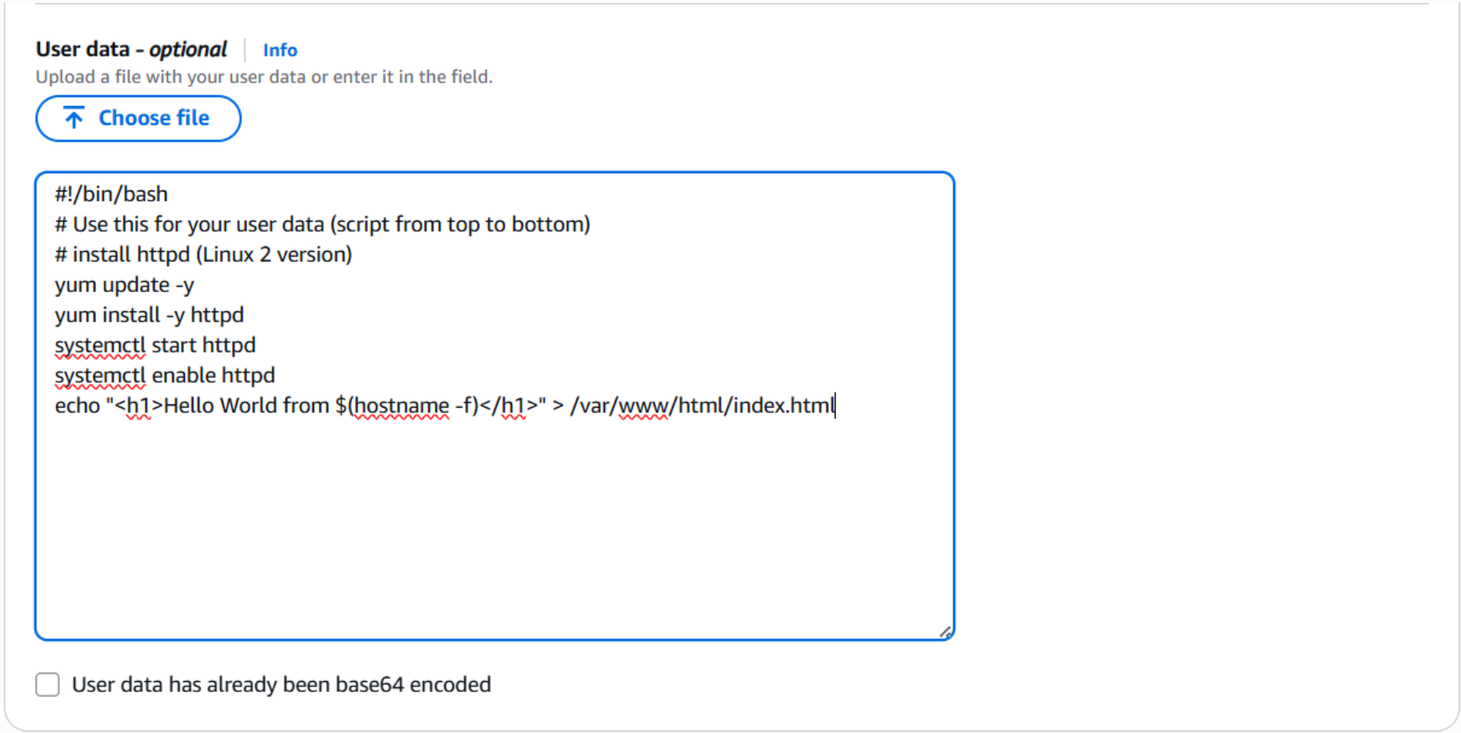
9. For the Security Group, use the same one from the **“AWS-LAB-01-EC2“**.



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Create Launch Template

10. Expand the **Advanced details** section and scroll down to **EC2 User Data**.
11. Enter the following commands, and make sure to include the <h1> tag.

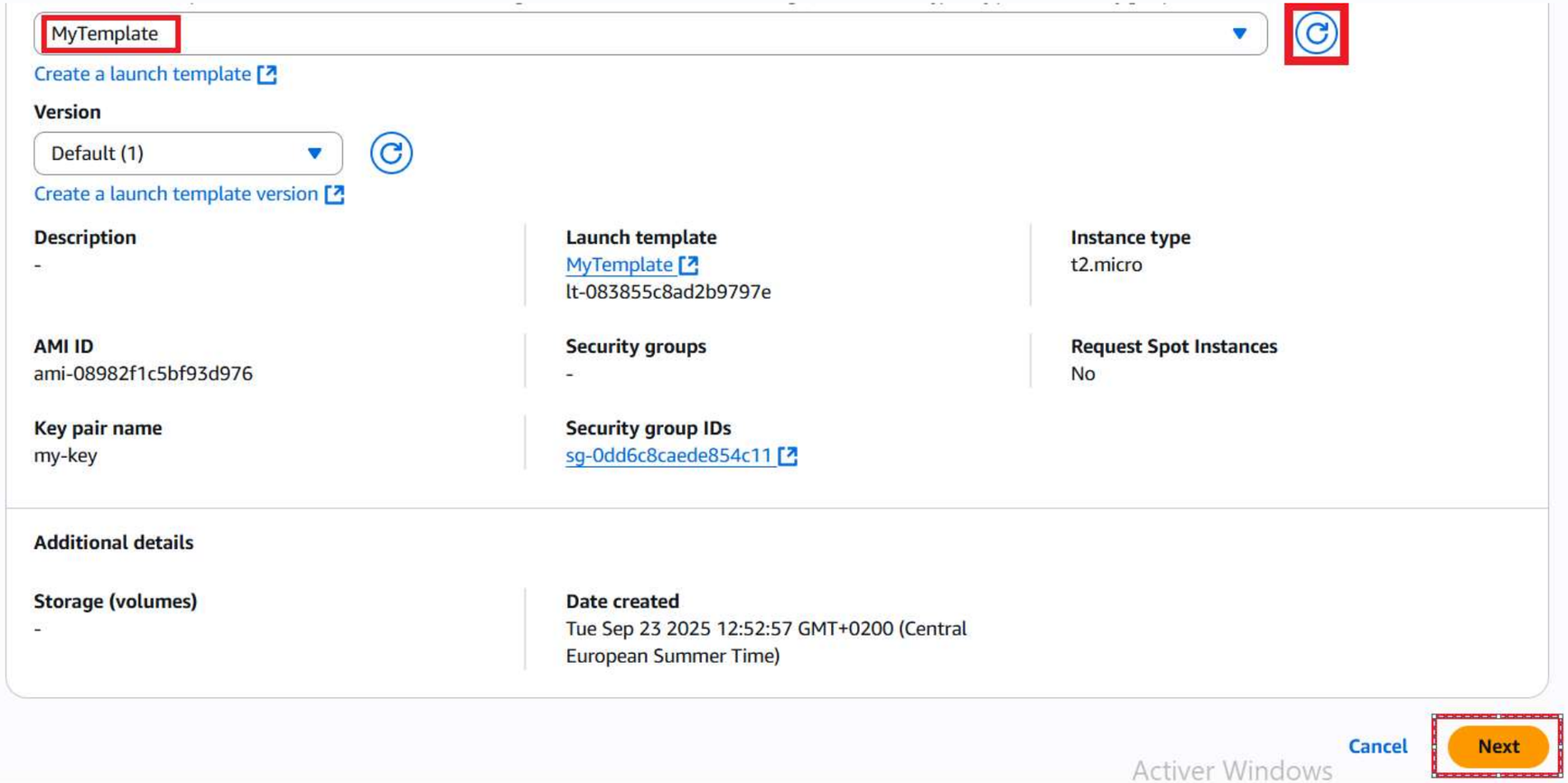


Then click on « **create launch template** »

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Create Auto-Scaling Group

13. Go back to the Create Auto Scaling Group page, refresh the list of launch templates, and select the one we just created, then **“Next”**

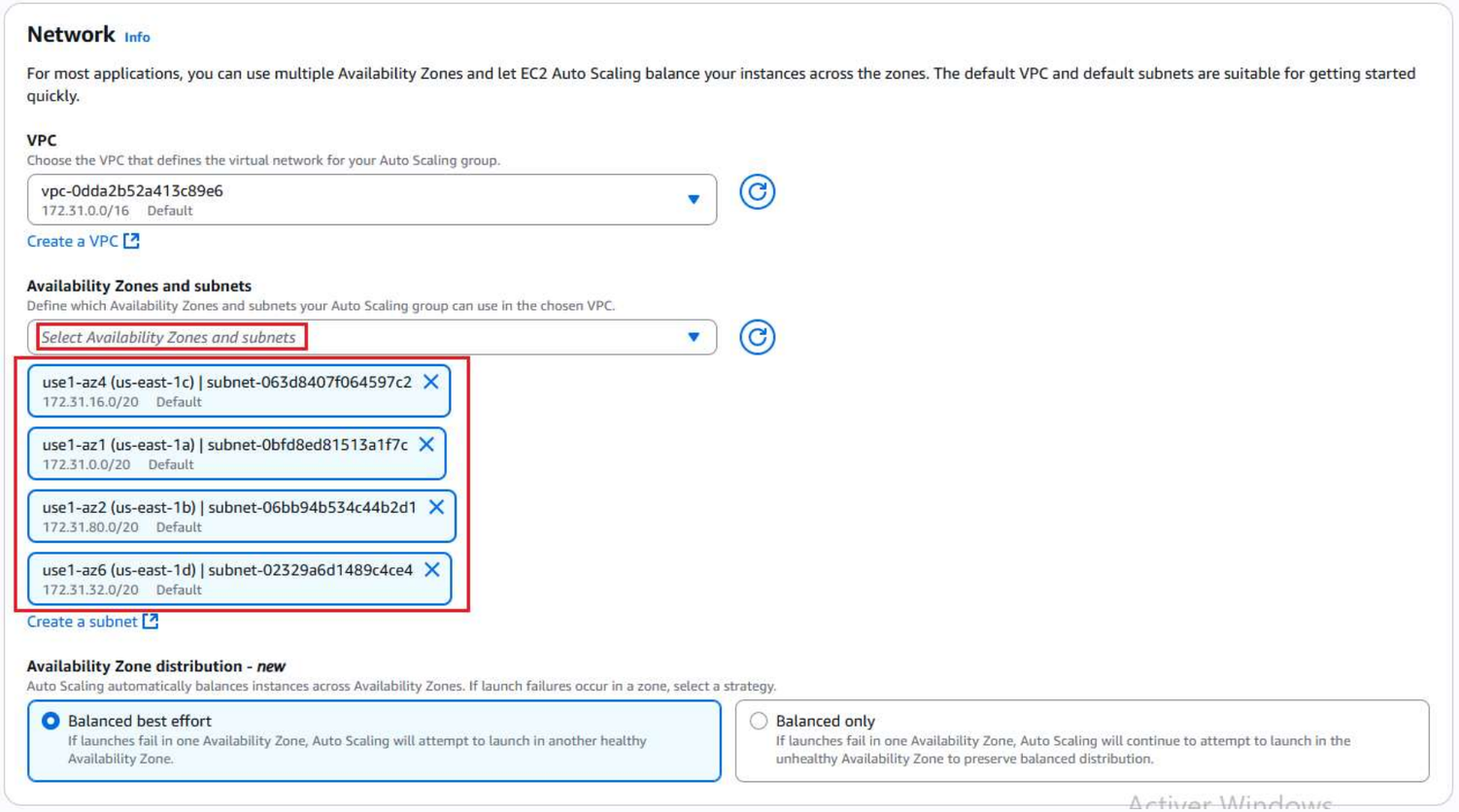


Activer Windows

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Create Auto-Scaling Group

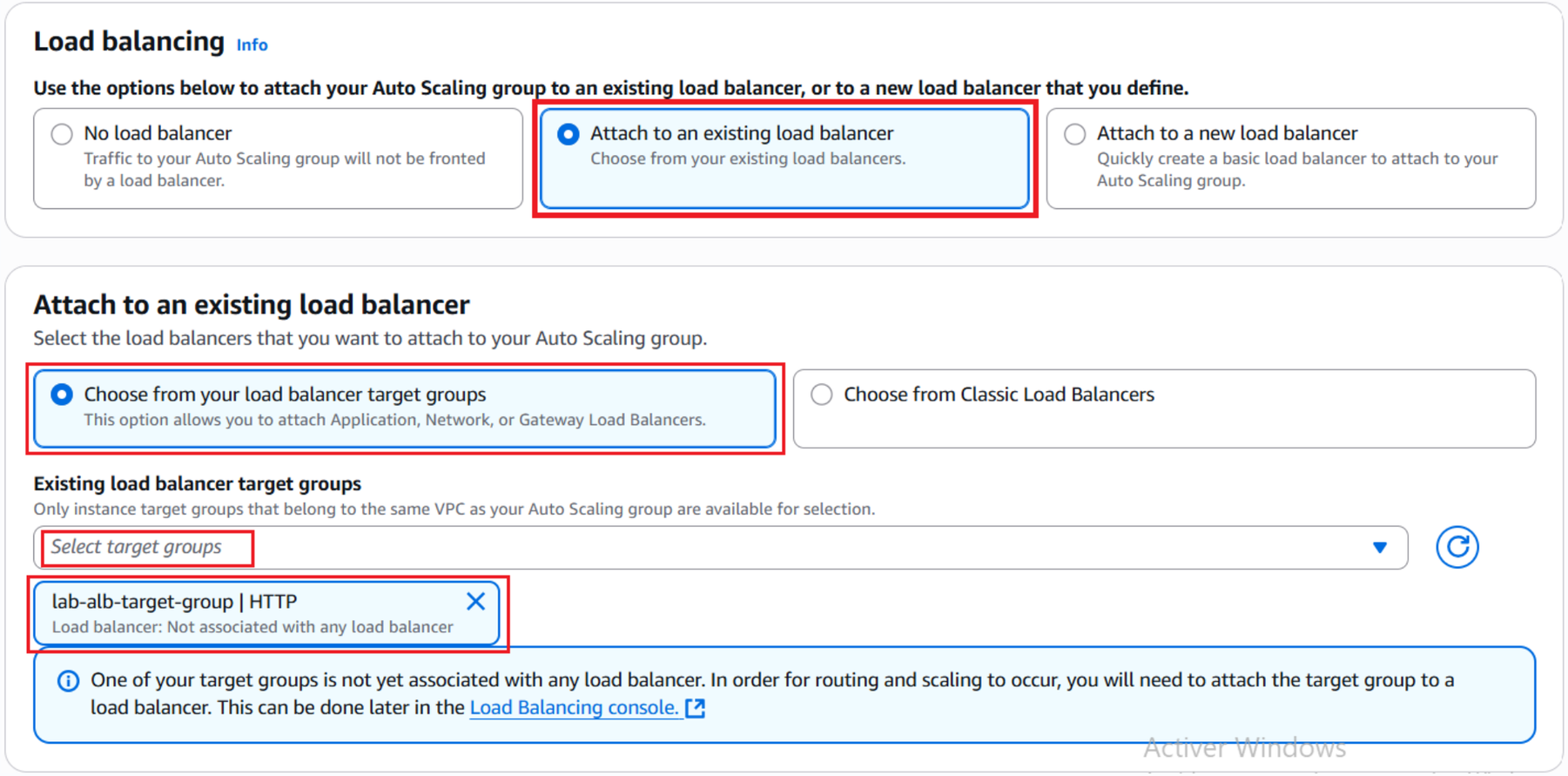
14. In the Network section Select 3 or 4 Availability Zones to make your Auto Scaling Group highly available



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Create Auto-Scaling Group

15. In the Load Balancer section, select “Attach to an existing load balancer” and choose the target group we created earlier..

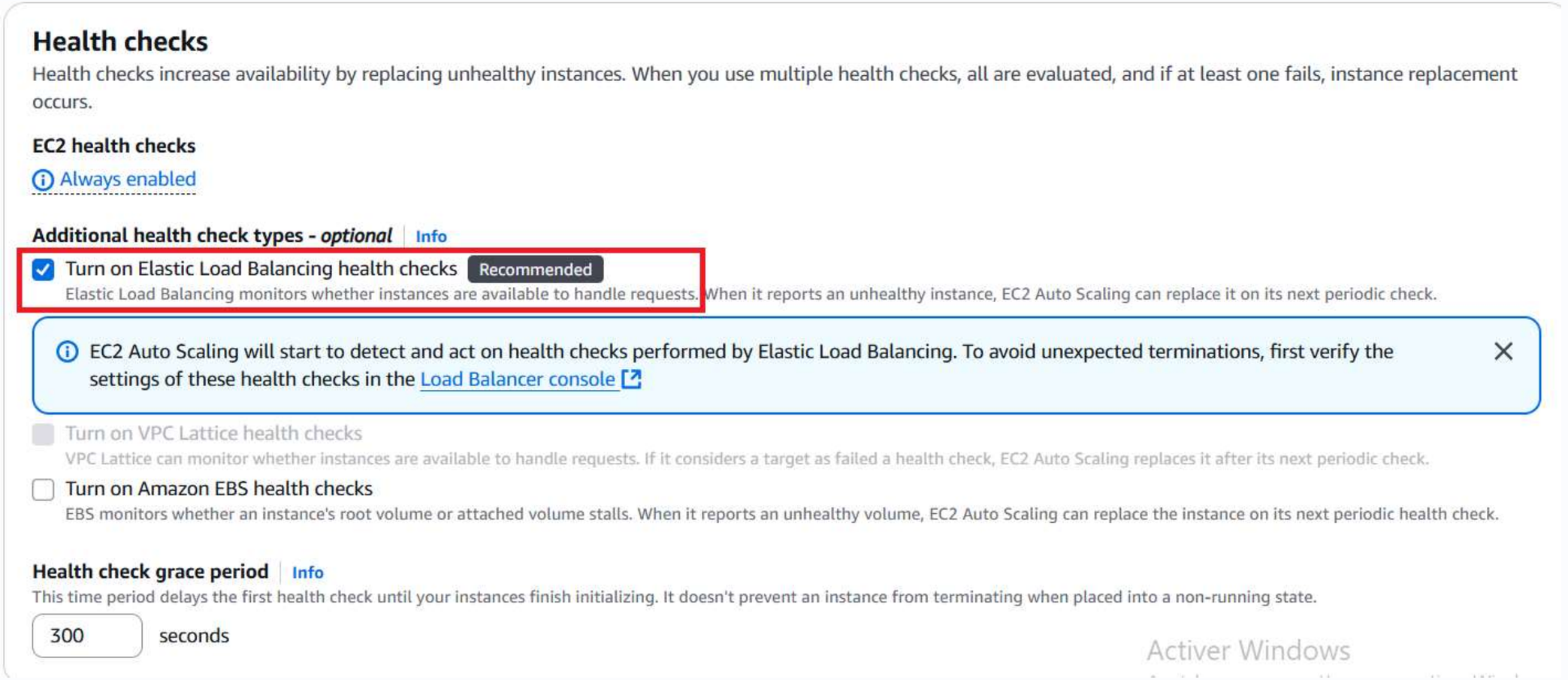


Activer Windows

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Create Auto-Scaling Group

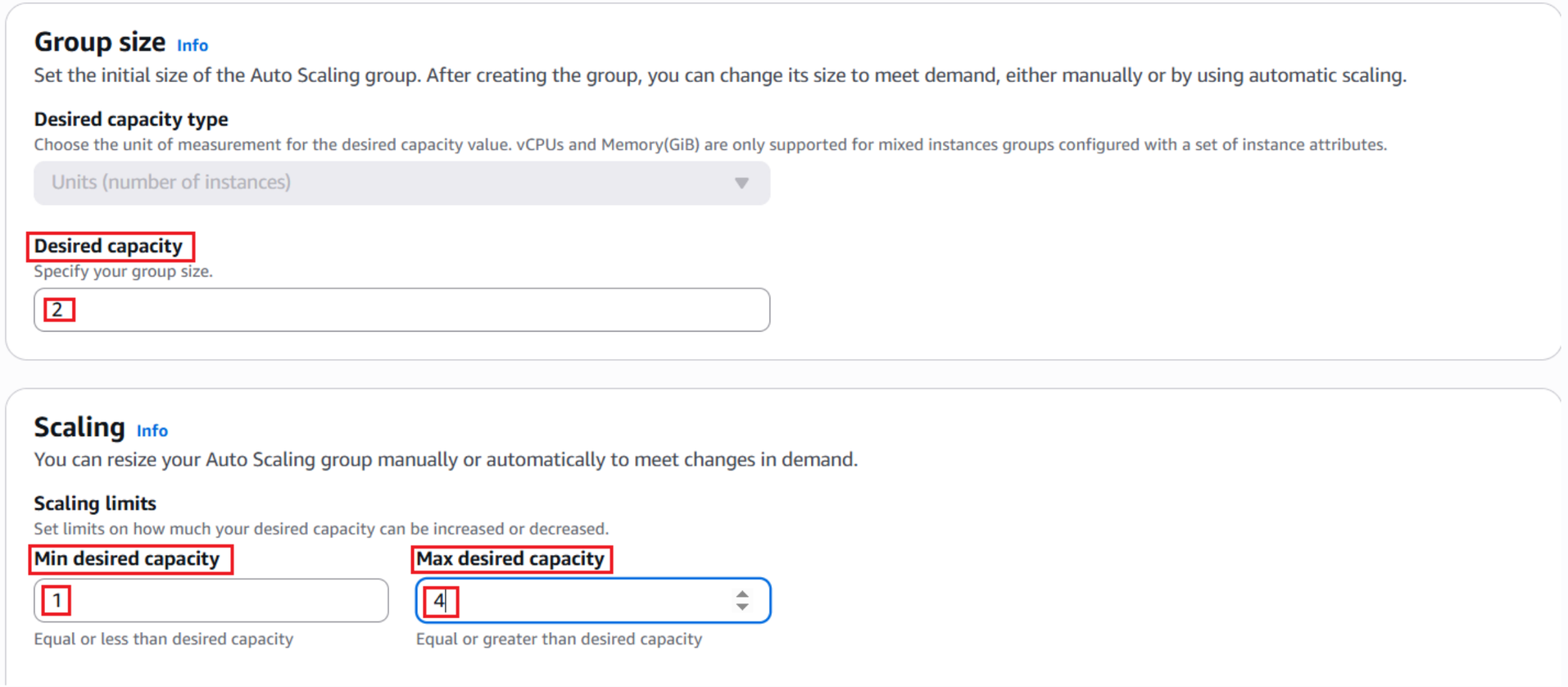
16. Enable the health check so the Auto Scaling Group can perform regular checks on the instances.



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Create Auto-Scaling Group

17. Set the desired capacity, minimum capacity, and maximum capacity for your Auto Scaling Group.



18. Then “next”, “next”, “next”, and **“create auto scaling group”**.

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